

Certalytic - Product & Engineering Overview

Document type: Product & engineering specification

Stack: Laravel 13 · Inertia.js (React 19) · Mistral AI (La Plateforme, EU) · Hetzner EU · Laravel Cashier (Stripe)

Status: Production MVP — integrity screening, supplementary candidate insights, async role exports, audio transcription tooling, marketing site, and legal/compliance surfaces shipped

1. Product vision

Certalytic is a **decision-support integrity layer** for recruitment. It sits on top of existing ATS workflows — it does not replace them.

After a candidate completes their interview loop, recruiters upload the CV and interview transcripts (single paste or up to three uploaded files merged automatically) plus optional LinkedIn/GitHub signals. Certalytic synthesizes these inputs into a single **Integrity Score** with evidence-based flags and follow-up prompts, plus supplementary **behaviour** and **personality** analyses for hiring-manager context.

Core question: How likely is this candidate's interview performance representative of their real-time, independent, unassisted ability?

Workflow

ATS: applications → CV filter → interview loop conducted

↓

Certalytic: CV + merged interview transcripts (+ optional LinkedIn/GitHub) → Integrity Score + supplementary insights

Recruiters wait until the candidate is at the finish line, paste or transcribe relevant interview content, and receive an integrity breakdown with follow-up prompts. **One screening = one candidate screening** (CV + merged transcripts + scoring). One Mistral evaluation call per screening.

A separate **audio transcription tool** converts interview recordings (MP3/M4A/WAV) into speaker-labelled transcripts that recruiters can paste into screenings or review standalone.

Decision support disclaimer

Every score view and export displays:

This score represents a probability heuristic, not an absolute verdict. Use it to guide your human follow-up questions.

Language avoids "pass/fail" or "cheating detected." Prefer: *integrity indicators, signal density, follow-up suggested, inconsistency flagged*.

2. Product scope

Shipped capabilities

Pillar	Description
CV intelligence	Mistral OCR (<code>mistral-ocr-latest</code>) extracts PDF/DOCX CVs; <code>mistral-small-latest</code> evaluates timeline consistency, AI-text patterns, and role fit
Merged interview intelligence	Single manual transcript paste or up to three uploaded files merged server-side — Mistral infers virtual interview segments, scores each sequentially, and detects stylistic shifts between technical and behavioural sections
Supplementary candidate insights	Behaviour analysis (communication style, collaboration indicators, watchpoints) and personality analysis (work style, motivation signals, culture-fit indicators) — included in UI and PDF exports, excluded from the Integrity Score
Cross-source validation	Starter+: manual LinkedIn paste/URL and GitHub URL; GitHub public API enrichment when username resolved; Growth+ tier flag for full cross-source scoring path
Role context	Saved Role Profiles (all tiers); targeted scan assets on Scale+ (up to 3 OCR'd documents per role)
PDF exports	Per-candidate integrity dossier (sync download); per-role batch dossier (async queue job, watermarked, auto-download when ready)
Bulk upload	CSV and multi-file import for interview-stage candidates (all tiers)
Audio transcription	Standalone tool: Mistral Voxtral (<code>voxtral-mini-latest</code>) diarized transcription, speaker rename, paginated library, audio deleted after processing
Team workspaces	Multi-team model, invitations, seat-based plans, Stripe billing via Cashier

Pillar	Description
Marketing site	Public landing page (/) with hero, EU privacy, stats, How it works (3-step process with screenshot placeholders), product bento, demo preview, pricing (EUR), reviews, roadmap
Legal & compliance	Privacy Policy, Terms, DPA, Cookie Policy, Imprint (/legal/*); registration requires Terms + Privacy + DPA consent

Out of scope (current release)

- Real-time audio/video proctoring during live interviews
- Direct ATS integrations, public API, webhooks, SSO (roadmap — see §12)
- Automatic LinkedIn profile fetch (URL/text passed to model; fetch stub logs and defers to manual paste)
- Native Zoom / Teams / Meet import connectors (audio file upload supported today)
- Fake or mock evaluation paths — all screenings use live Mistral inference

3. Architecture

Component	Technology
Application	Laravel 13 + Inertia.js (React 19)
Document OCR	mistral-ocr-latest
Scoring & analysis	mistral-small-latest
Audio transcription	voxtral-mini-latest (Mistral La Plateforme)
PDF generation	DomPDF (watermarked dossiers)
Object storage	Hetzner Object Storage (EU, S3-compatible)
Queue	Laravel Queue (Redis) — screenings, priority screenings, transcriptions, role PDF exports
Database	PostgreSQL
Compute	Hetzner Cloud + Coolify (EU)
Billing	Laravel Cashier + Stripe (billing only — no candidate PII in payments path)

All document and audio intelligence routes to **Mistral La Plateforme (Paris, France)**. No US AI providers in the candidate data path.

Screening processing pipeline

1. Recruiter submits candidate: CV (upload or paste) + interview transcript (single paste or up to 3 files merged) + optional profile URLs; audio can be transcribed separately via Tools → Transcription
2. Record created; `ProcessCandidateScreeningJob` dispatched to Redis queue (priority queue on Scale+)
3. CV resolved (stored text, local DOCX/MD reader, or Mistral OCR for PDF)
4. GitHub profile fetched via public API when URL/username provided and text not already pasted
5. Mistral Small evaluates CV + merged transcript + profiles with role context (title, JD, scan assets); returns integrity components, virtual `round_analyses`, and supplementary `behaviour_analysis` + `personality_analysis`
6. Integrity Score composed locally from the four weighted components only; virtual segment scores persisted; UI updates when job completes

Role export pipeline

1. Recruiter requests export from role detail page or roles index dropdown
2. `role_exports` record created; `GenerateRoleExportPdfJob` queued
3. Worker loads role + all completed candidates, renders combined watermarked PDF (role overview + per-candidate dossiers including supplementary insights)
4. File stored in object storage; frontend polls and auto-downloads when complete

Transcription processing pipeline

1. Recruiter uploads audio (MP3/M4A/WAV) on **Tools → Transcription**; one **transcription token** consumed per job
2. `TranscribeAudioJob` dispatched to `transcriptions` queue
3. Mistral Voxtral returns diarized segments; formatted as `Speaker: line` transcript with `segments` + `speaker_labels` persisted
4. Source audio file deleted from storage after successful processing (`audio_path` nulled)
5. Recruiter can rename speakers, search transcripts, copy text into a screening, or delete records

Transcript controls (screening input)

Mode	Control	Value
Manual paste	Fields	1 (single textarea)
File upload	Max files	3 (merged server-side with segment separators)
Validation (upload/paste)	Max words (combined)	20,000
Validation	Max characters (combined)	150,000
Processing (TranscriptProcessor)	Soft warning	~24,000 words
Processing	Hard cap	120,000 characters per merged transcript
Processing	Truncation	Preserves questions, answers, intro/outro; flags was_truncated on report

Audio upload limits (transcription tool)

Control	Default
Max file size	100 MB (CERTALYTIC_AUDIO_MAX_KB)
Max duration	60 minutes
Formats	MP3, M4A, WAV

4. Scoring engine

Integrity Score = (S_{cv} × 0.25) + (S_{int} × 0.50) + (S_{cross} × 0.15) + (S_{id} × 0.10)

Component	Weight	Source
\$S_{cv}\$	25%	CV authenticity & role fit
\$S_{int}\$	50%	Interview signal density (weighted across virtual segments when multiple are inferred)
\$S_{cross}\$	15%	CV vs LinkedIn/GitHub consistency

Component	Weight	Source
<code>\$S_{id}\$</code>	10%	Identity & communication coherence

Not included in the Integrity Score: behaviour analysis and personality analysis. These supplementary sections are generated in the same Mistral call but stored separately in `score_breakdown` and surfaced only in dedicated UI tabs and PDF export sections.

Mistral returns schema-validated JSON including `round_analyses` for inferred interview segments, plus `behaviour_analysis` and `personality_analysis`. Laravel applies weights deterministically to evaluated integrity components only. When no LinkedIn or GitHub profiles are submitted, `s_cross` is stored as `not-evaluated` and excluded from the integrity score; the remaining component weights are renormalized to 100%. The full merged transcript is stored in one `interview_rounds` row (`round_number = 1`); additional virtual segments are tracked in `round_analyses` and synced to supplementary `interview_rounds` rows for reporting.

Virtual segment weighting

When Mistral identifies multiple segments, `$S_{int}$` uses weighted rolling scores:

Segment	Weight
1	25%
2	35%
3	40%

Variance between consecutive segment scores flags high inconsistency (threshold: 20 points).

Supplementary analyses (outside integrity score)

Analysis	Contents
Behaviour	Summary, observed traits, communication style, collaboration indicators, watchpoints
Personality	Summary, observed traits, work style, motivation signals, culture-fit indicators

Displayed on candidate detail tabs and included in single-candidate and role-batch PDF exports with an explicit disclaimer that they do not affect the Integrity Score.

5. Role profiles

Roles are **title + job description only** — no per-role interview round configuration.

Tier	Roles
All tiers	Saved Role Profiles (shared org library, Roles tab, dashboard filtering)
Scale+	Up to 3 targeted scan assets per role (take-home instructions, rubrics) — OCR'd once, cached, cross-referenced in merged transcript analysis

6. Plans & pricing (EUR)

All public and in-app pricing is displayed in **euros (€)**.

Monthly plans

One screening = one candidate screening (CV + merged transcripts + integrity scoring + supplementary behaviour/personality insights).

Plan	Price/mo	Seats	Screenings/mo	Key gates
Free	€0	1	3	Summary score only, watermarked exports
Starter	€159	1	20	Full breakdown, manual LinkedIn/GitHub URLs, behaviour & personality insights
Growth	€349	3	50	Cross-source tier flag, 3 seats
Scale	€799	5	125	Priority queue, role scan assets
Enterprise	Custom	6+	Custom	Sales-led; ATS / API / SSO on roadmap

Screening packs (paid plans)

One-off screening credits. Packs expire at billing cycle end.

Pack	Price	Screenings	Per screening
Quick Refill	€99	10	€9.90
Pipeline Surge	€299	35	€8.50
High-Volume Boost	€750	100	€7.50

Transcription tokens

Separate from screening quota. Consumed by the standalone audio transcription tool (1 token = 1 audio job).

Pack	Price	Tokens
Transcription Token Pack	€15	5

Purchasable from Tools → Transcription via Stripe Checkout. Credited to `teams.transcript_tokens`.

Feature matrix (selected)

Capability	Free	Starter	Growth	Scale
CV + merged interview scoring	✓	✓	✓	✓
Full integrity score breakdown	—	✓	✓	✓
Behaviour & personality analysis	✓	✓	✓	✓
Saved Role Profiles	✓	✓	✓	✓
Cross-source (LinkedIn/GitHub)	—	Manual URLs / paste	✓	✓
Targeted scan assets	—	—	—	✓ (max 3)
Priority screening queue	—	—	—	✓
Per-candidate PDF export	✓*	✓	✓	✓
Async role batch PDF export	✓*	✓	✓	✓
Screening packs	—	✓	✓	✓
Speaker-labelled audio transcription	✓**	✓**	✓**	✓**
EU sovereign processing	✓	✓	✓	✓

*Free tier exports are watermarked.

**Requires transcription tokens (pack purchase); not included in monthly screening quota.

7. Marketing site & public surfaces

Public route: `/` (welcome Inertia page).

Section	Purpose
---------	---------

Hero	Value prop (€100k mistake framing), CTA (register / dashboard), Contact mailto
EU privacy	GDPR-first infrastructure, DATA & PRIVACY panel, Hetzner/Mistral/Stripe chips
Stats	Configurable marketing metrics (candidates screened, customers, countries, audio hours, cost avoided)
How it works	3-step process: create role → screen candidates (multi-step form screenshots) → export report
Product bento	Four integrity signals + platform features including behaviour/personality analysis
Demo	Full mocked screening result preview with supplementary insights
Pricing	Starter, Growth (highlighted), Scale, Enterprise — EUR pricing; Free tier mentioned in copy
Reviews	Testimonials with star ratings and avatar initials
Roadmap	Quarterly planned features (synced with §12)
CTA	Three free screenings

Marketing stats are configurable via `CERTALYTIC_MARKETING_*` env vars (see §11).

8. Data & compliance

- **EU-only path:** Hetzner datacenters (DE/FI) + Mistral La Plateforme (Paris, FR). No US cloud compute or CDN in the candidate data path.
- **Ephemeral processing:** CV bytes sent to Mistral OCR per request; not retained on Mistral infrastructure.
- **Audio retention:** Interview audio deleted from object storage after transcription completes.
- **No training:** Candidate data is never used to train, fine-tune, or improve Mistral foundation models (contractual API tier + surfaced in UI DATA & PRIVACY panel).
- **Sub-processors:** Hetzner Online GmbH (EU), Mistral AI (EU), Stripe (billing metadata only).
- **Legal documents:** Privacy Policy, Terms of Service, Data Processing Agreement, Cookie Policy, Imprint — linked from footer and registration flow.

- **Registration consent:** Users must accept Terms, Privacy Policy, and DPA at signup (`accept_terms` , `accept_privacy` , `accept_dpa`).

Company and social links are configured via `CERTALYTIC_COMPANY_*` and `CERTALYTIC_SOCIAL_*` env vars, shared globally through Inertia.

9. Key database entities

- `users` — authentication (Fortify), current team selection
- `teams` — org workspace, plan, Stripe billing (Cashier), `transcript_tokens` balance
- `team_user` — membership, role (owner/member)
- `team_invitations` — email invitations with accept flow
- `roles` — title, description (all tiers with saved roles)
- `role_documents` — scan assets (Scale+)
- `role_exports` — async role PDF export jobs (status, path, error)
- `candidates` — screening record, integrity score, score breakdown (integrity components + supplementary analyses + virtual `round_analyses`)
- `interview_rounds` — merged transcript in round 1; supplementary rows for virtual segments 2+; optional audio fields for in-flow transcription
- `audio_transcriptions` — standalone transcription jobs: status, diarized `segments` , `speaker_labels` , formatted `transcript_text`
- `transcript_token_credits` — idempotent Stripe checkout credit ledger

10. Environment

```
# Mistral (required – no fake evaluator fallback)
MISTRAL_API_KEY=
MISTRAL_CHAT_MODEL=mistral-small-latest
MISTRAL_OCR_MODEL=mistral-ocr-latest
MISTRAL_TRANSCRIPTION_MODEL=voxtral-mini-latest

# Queue (required for async screening + transcription + role export jobs)
QUEUE_CONNECTION=redis
CERTALYTIC_QUEUE=default
CERTALYTIC_PRIORITY_QUEUE=screenings-priority
CERTALYTIC_TRANSCRIPTIONS_QUEUE=transcriptions
```

```

# Stripe / Cashier (billing + screening packs)
STRIPE_KEY=
STRIPE_SECRET=
STRIPE_WEBHOOK_SECRET=
STRIPE_PRICE_STARTER=
STRIPE_PRICE_GROWTH=
STRIPE_PRICE_SCALE=
STRIPE_PRICE_PACK_QUICK=
STRIPE_PRICE_PACK_SURGE=
STRIPE_PRICE_PACK_BOOST=
STRIPE_PRICE_TRANSCRIPT_FIVE_PACK=

# Company / legal (shared to Inertia as company + socialLinks)
CERTALYTIC_COMPANY_EMAIL=hello@certalytic.com
CERTALYTIC_COMPANY_LEGAL_NAME="Certalytic GmbH"
# ... see .env.example for full list

# Marketing stats (landing page)
CERTALYTIC_MARKETING_CANDIDATES_SCREENED="12,400+"
CERTALYTIC_MARKETING_CUSTOMERS="180+"
CERTALYTIC_MARKETING_COUNTRIES="14"
CERTALYTIC_MARKETING_AUDIO_HOURS="2,800+"
CERTALYTIC_MARKETING_SAVED_MILLIONS="€4.2M"

```

Run `php artisan queue:work` (or `composer run dev`) so screenings, transcriptions, and role PDF exports process after submission.

11. Roadmap

Approximate delivery quarters (also rendered on the public landing page). Items not yet built.

Quarter	Feature	Description
Q3 2026	ATS integrations	Push integrity reports into Greenhouse, Lever, and Workday without manual copy-paste
Q4 2026	Enterprise SSO	SAML and OIDC single sign-on with seat provisioning for agency and in-house TA teams
Q1 2027	Batch screening	Upload a cohort of candidates against one role profile and compare integrity signals side by side

Quarter	Feature	Description
Q2 2027	Public API & webhooks	Programmatic screening triggers and signed webhook delivery for custom hiring stacks

Additional backlog (unscheduled)

- Automatic LinkedIn profile fetch API (today: manual paste or URL-only passthrough)
- Native Zoom / Teams / Google Meet recording import
- Re-screening with component-level cache reuse
- Annual billing option
- Enterprise: dedicated success manager, custom onboarding (sales-led today)

Last aligned with codebase: June 2026.